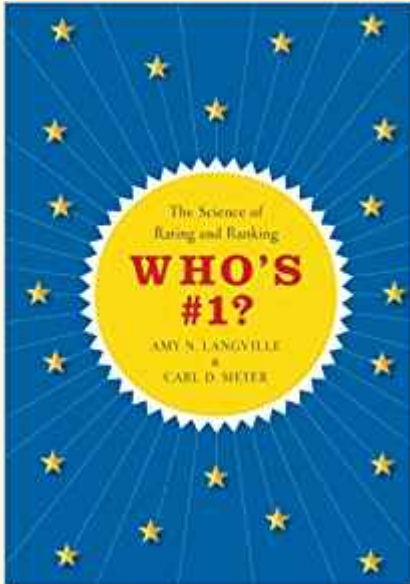


RANKING AND RATING: MARKOV CHAINS

AP

Ch. 6 of Langville-Meyer's textbook is dedicated to Markov chains in sport prediction

We learn a new key concept of Data Science



NOTATION

A STOCHASTIC MATRIX S

describes the probab. of a *transition* of some sort between places or states etc.

$$s_{ij} = Pr[\text{the system goes from } i \text{ to } j]$$

As a result:

$$\mathbf{S} = \begin{matrix} & \begin{matrix} \text{Duke} & \text{Miami} & \text{UNC} & \text{UVA} & \text{VT} \end{matrix} \\ \begin{matrix} \text{Duke} \\ \text{Miami} \\ \text{UNC} \\ \text{UVA} \\ \text{VT} \end{matrix} & \left(\begin{array}{ccccc} 0 & 1/4 & 1/4 & 1/4 & 1/4 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \\ 0 & 1/2 & 0 & 0 & 1/2 \\ 0 & 1/3 & 1/3 & 0 & 1/3 \\ 0 & 1 & 0 & 0 & 0 \end{array} \right) \end{matrix}$$

NOTATION OF THE CHAPTER

Notation for the Markov Rating Method

- k number of statistics to incorporate into the Markov model
- $V_{stat1}, V_{stat2}, \dots, V_{statk}$ raw voting matrix for each game statistic k
 $[V_{stat}]_{ij}$ = number of votes team i casts for team j using statistic $stat$
- $S_{stat1}, S_{stat2}, \dots, S_{statk}$ stochastic matrices built from corresponding voting matrices $V_{stat1}, V_{stat2}, \dots, V_{statk}$
- S final stochastic matrix built from $S_{stat1}, S_{stat2}, \dots, S_{statk}$;
 $S = \alpha_1 S_{stat1} + \alpha_2 S_{stat2} + \dots + \alpha_k S_{statk}$
- α_i weight associated with game statistic i ; $\sum_{i=1}^k \alpha_i = 1$ and $\alpha_i \geq 0$.
- $\bar{S}$ stochastic Markov matrix that is guaranteed to be irreducible;
 $\bar{S} = \beta S + (1 - \beta)/n \mathbf{E}$, $0 < \beta < 1$
- r Markov rating vector; stationary vector (i.e., dominant eigenvector) of $\bar{S}$
- n number of teams in the league = order of $\bar{S}$

THE MARKOV METHOD

THE FAIRWEATHER FAN

switches their allegiance to the winning team **of the moment**.

If they support i , what is the prob. that they switch to j ?

$$\mathbf{S} = \begin{matrix} & \begin{matrix} \text{Duke} & \text{Miami} & \text{UNC} & \text{UVA} & \text{VT} \end{matrix} \\ \begin{matrix} \text{Duke} \\ \text{Miami} \\ \text{UNC} \\ \text{UVA} \\ \text{VT} \end{matrix} & \left(\begin{array}{ccccc} 0 & 1/4 & 1/4 & 1/4 & 1/4 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \\ 0 & 1/2 & 0 & 0 & 1/2 \\ 0 & 1/3 & 1/3 & 0 & 1/3 \\ 0 & 1 & 0 & 0 & 0 \end{array} \right) \end{matrix}$$

How did we obtain this matrix?

Input: the win-loss data:

$$\mathbf{V} = \begin{array}{c} \text{Duke} \\ \text{Miami} \\ \text{UNC} \\ \text{UVA} \\ \text{VT} \end{array} \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{pmatrix}$$

Rows normalised to 1:

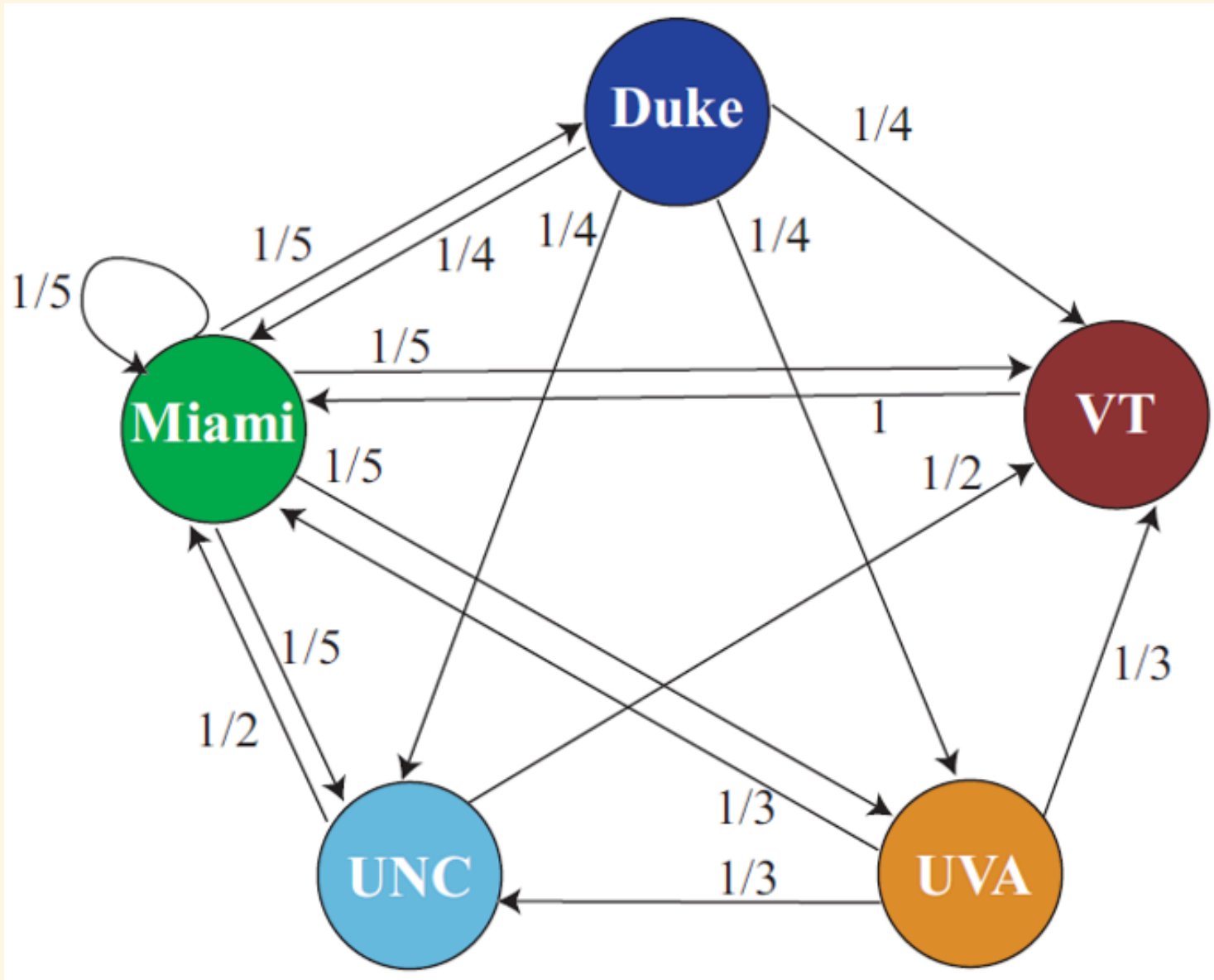
$$\mathbf{N} = \begin{matrix} & \begin{matrix} \text{Duke} & \text{Miami} & \text{UNC} & \text{UVA} & \text{VT} \end{matrix} \\ \begin{matrix} \text{Duke} \\ \text{Miami} \\ \text{UNC} \\ \text{UVA} \\ \text{VT} \end{matrix} & \begin{pmatrix} 0 & 1/4 & 1/4 & 1/4 & 1/4 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 & 1/2 \\ 0 & 1/3 & 1/3 & 0 & 1/3 \\ 0 & 1 & 0 & 0 & 0 \end{pmatrix} \end{matrix}$$

The Miami row sums to 0: not stochastic!

As with PageRank, substitute all $\mathbf{0}^T$ rows with $\frac{1}{n} \mathbf{1}^T$

$$\mathbf{S} = \begin{matrix} & \begin{matrix} \text{Duke} & \text{Miami} & \text{UNC} & \text{UVA} & \text{VT} \end{matrix} \\ \begin{matrix} \text{Duke} \\ \text{Miami} \\ \text{UNC} \\ \text{UVA} \\ \text{VT} \end{matrix} & \left(\begin{array}{ccccc} 0 & 1/4 & 1/4 & 1/4 & 1/4 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \\ 0 & 1/2 & 0 & 0 & 1/2 \\ 0 & 1/3 & 1/3 & 0 & 1/3 \\ 0 & 1 & 0 & 0 & 0 \end{array} \right) \end{matrix}$$

Now the fair-weather fan takes a long, random walk along this *Markov graph*:



We record the number of times the random walker passes each vertex.

After a while, the proportion of visits to each node stabilizes.

The vector $\mathbf{r}$ with the frequencies is a *stationary vector*

$\mathbf{r}$ corresponds to the dominant e-vector of the Markov-chain matrix!

Team	$\mathbf{r}$	Rank
Duke	.087	5th
Miami	.438	1st
UNC	.146	3rd
UVA	.110	4th
VT	.219	2nd

HOW TO CREATE THE BASE MATRIX

WITH POINTS DIFFERENTIAL

$$\mathbf{V} = \begin{matrix} & \begin{matrix} \text{Duke} & \text{Miami} & \text{UNC} & \text{UVA} & \text{VT} \end{matrix} \\ \begin{matrix} \text{Duke} \\ \text{Miami} \\ \text{UNC} \\ \text{UVA} \\ \text{VT} \end{matrix} & \left(\begin{array}{ccccc} 0 & 45 & 3 & 31 & 45 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 18 & 0 & 0 & 27 \\ 0 & 8 & 2 & 0 & 38 \\ 0 & 20 & 0 & 0 & 0 \end{array} \right) \end{matrix}$$

$$\mathbf{S} = \begin{matrix} & \begin{matrix} \text{Duke} & \text{Miami} & \text{UNC} & \text{UVA} & \text{VT} \end{matrix} \\ \begin{matrix} \text{Duke} \\ \text{Miami} \\ \text{UNC} \\ \text{UVA} \\ \text{VT} \end{matrix} & \left(\begin{array}{ccccc} 0 & 45/124 & 3/124 & 31/124 & 45/124 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \\ 0 & 18/45 & 0 & 0 & 27/45 \\ 0 & 8/48 & 2/48 & 0 & 38/48 \\ 0 & 1 & 0 & 0 & 0 \end{array} \right) \end{matrix}$$

Team	r	Rank
Duke	.088	5th
Miami	.442	1st
UNC	.095	4th
UVA	.110	3rd
VT	.265	2nd

WINNERS/LOSERS WITH POINTS

$$S_{point} = \begin{matrix} & \begin{matrix} \text{Duke} & \text{Miami} & \text{UNC} & \text{UVA} & \text{VT} \end{matrix} \\ \begin{matrix} \text{Duke} \\ \text{Miami} \\ \text{UNC} \\ \text{UVA} \\ \text{VT} \end{matrix} & \left(\begin{array}{ccccc} 0 & 52/159 & 24/159 & 38/159 & 45/159 \\ 7/47 & 0 & 16/47 & 17/47 & 7/47 \\ 21/90 & 34/90 & 0 & 5/90 & 30/90 \\ 7/91 & 25/91 & 7/91 & 0 & 52/91 \\ 0 & 27/44 & 3/44 & 14/44 & 0 \end{array} \right) \end{matrix}$$

$$\mathbf{S}_{point} = \begin{matrix} & \begin{matrix} \text{Duke} & \text{Miami} & \text{UNC} & \text{UVA} & \text{VT} \end{matrix} \\ \begin{matrix} \text{Duke} \\ \text{Miami} \\ \text{UNC} \\ \text{UVA} \\ \text{VT} \end{matrix} & \left(\begin{array}{ccccc} 0 & 52/159 & 24/159 & 38/159 & 45/159 \\ 7/47 & 0 & 16/47 & 17/47 & 7/47 \\ 21/90 & 34/90 & 0 & 5/90 & 30/90 \\ 7/91 & 25/91 & 7/91 & 0 & 52/91 \\ 0 & 27/44 & 3/44 & 14/44 & 0 \end{array} \right) \end{matrix}$$

Team	r	Rank
Duke	.095	5th
Miami	.296	1st
UNC	.149	4th
UVA	.216	3rd
VT	.244	2nd

WITH YARDAGE

	Duke	Miami	UNC	UVA	VT
Duke		100-557	209-357	207-315	35-362
Miami	557-100		321-188	500-452	304-167
UNC	357-209	188-321		270-199	196-338
UVA	315-207	452-500	199-270		334-552
VT	362-35	167-304	338-196	552-334	

$$\mathbf{V}_{yardage} = \begin{matrix} & \begin{matrix} \text{Duke} & \text{Miami} & \text{UNC} & \text{UVA} & \text{VT} \end{matrix} \\ \begin{matrix} \text{Duke} \\ \text{Miami} \\ \text{UNC} \\ \text{UVA} \\ \text{VT} \end{matrix} & \begin{pmatrix} 0 & 577 & 357 & 315 & 362 \\ 100 & 0 & 188 & 452 & 167 \\ 209 & 321 & 0 & 199 & 338 \\ 207 & 500 & 270 & 0 & 552 \\ 35 & 304 & 196 & 334 & 0 \end{pmatrix} \end{matrix}$$

$$\mathbf{S}_{yardage} = \begin{matrix} & \begin{matrix} \text{Duke} & \text{Miami} & \text{UNC} & \text{UVA} & \text{VT} \end{matrix} \\ \begin{matrix} \text{Duke} \\ \text{Miami} \\ \text{UNC} \\ \text{UVA} \\ \text{VT} \end{matrix} & \left(\begin{array}{ccccc} 0 & 577/1611 & 357/1611 & 315/1611 & 362/1611 \\ 100/907 & 0 & 188/907 & 452/907 & 167/907 \\ 209/1067 & 321/1067 & 0 & 199/1067 & 338/1067 \\ 207/1529 & 500/1529 & 270/1529 & 0 & 552/1529 \\ 35/869 & 304/869 & 196/869 & 334/869 & 0 \end{array} \right) \end{matrix}$$

Team	r	Rank
Duke	.105	5th
Miami	.249	2nd
UNC	.170	4th
UVA	.260	1st
VT	.216	3rd

WITH TURNOVER

$$\mathbf{S}_{turnover} = \begin{matrix} & \begin{matrix} \text{Duke} & \text{Miami} & \text{UNC} & \text{UVA} & \text{VT} \end{matrix} \\ \begin{matrix} \text{Duke} \\ \text{Miami} \\ \text{UNC} \\ \text{UVA} \\ \text{VT} \end{matrix} & \begin{pmatrix} 0 & 1/9 & 3/9 & 4/9 & 1/9 \\ 3/9 & 0 & 4/9 & 1/9 & 1/9 \\ 2/7 & 3/7 & 0 & 1/7 & 1/7 \\ 1/5 & 0 & 1/5 & 0 & 3/5 \\ 1/10 & 6/10 & 2/10 & 1/10 & 0 \end{pmatrix} \end{matrix}$$

WITH POSSESSION

$$\mathbf{S}_{poss} = \begin{matrix} & \begin{matrix} \text{Duke} & \text{Miami} & \text{UNC} & \text{UVA} & \text{VT} \end{matrix} \\ \begin{matrix} \text{Duke} \\ \text{Miami} \\ \text{UNC} \\ \text{UVA} \\ \text{VT} \end{matrix} & \left(\begin{array}{ccccc} 0 & 29.7/118.6 & 30.8/118.6 & 28/118.6 & 30.1/118.6 \\ 30.3/123.6 & 0 & 36.3/123.6 & 31/123.6 & 26/123.6 \\ 29.1/111.6 & 23.7/111.6 & 0 & 27.5/111.6 & 31.3/111.6 \\ 32/131.9 & 29/131.9 & 32.5/131.9 & 0 & 38.4/131.9 \\ 29.9/114.2 & 34/114.2 & 28.7/114.2 & 21.6/114.2 & 0 \end{array} \right) \end{matrix}$$

WITH LINEAR COMBINATIONS OF FEATURES

$$\mathbf{S} = \alpha_1 \mathbf{S}_{\text{points}} + \alpha_2 \mathbf{S}_{\text{yard.}} + \alpha_3 \mathbf{S}_{\text{turn.}} + \alpha_4 \mathbf{S}_{\text{poss.}}$$

If weights are all non-negative and sum to 1, also $\mathbf{S}$ will be stochastic.

Weights are assigned by experts or...

could be learned by an outer ML system running on historical data.

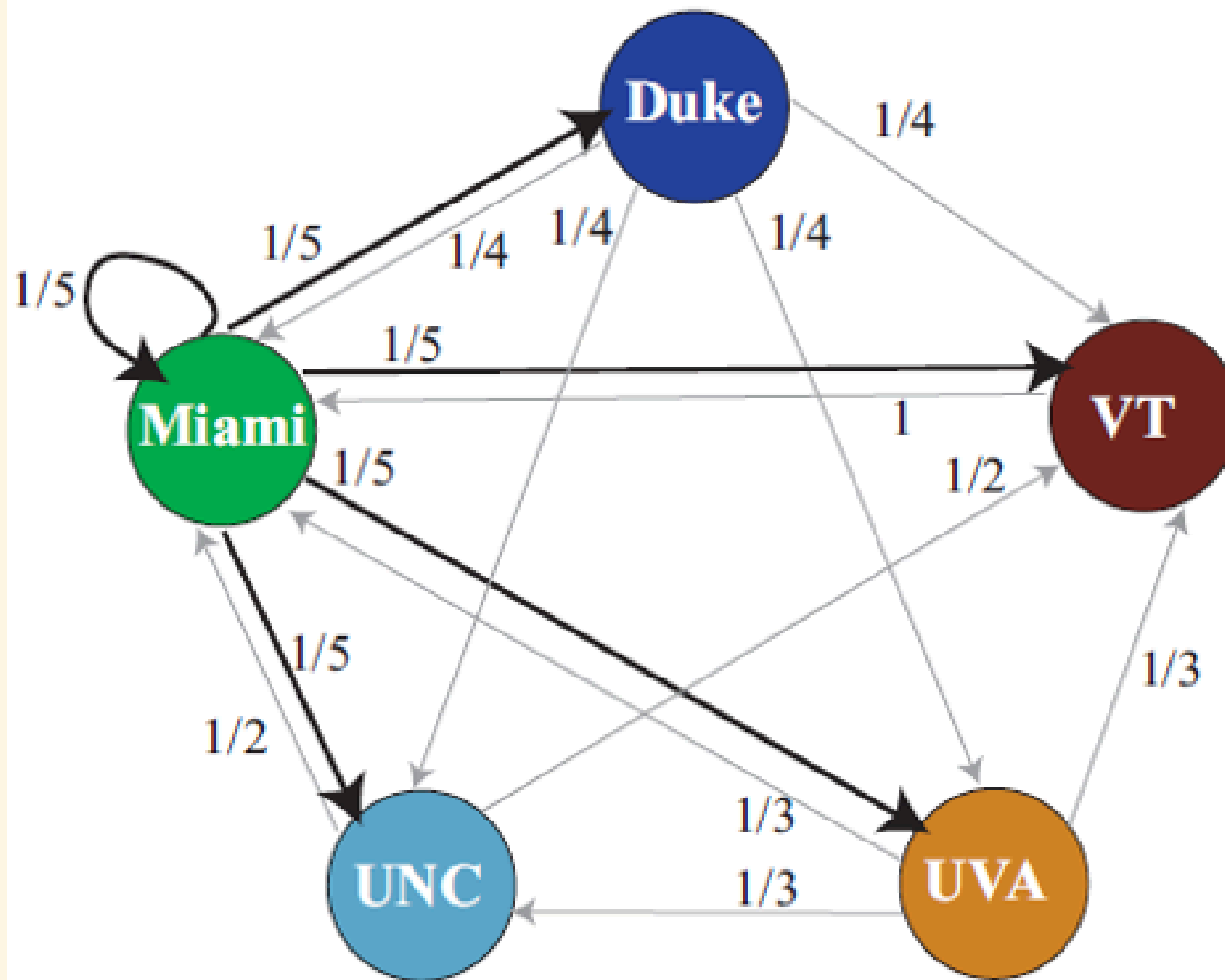
By default, let's set all 4 α weights to $\frac{1}{4}$:

$$\mathbf{S} = \begin{matrix} & \begin{matrix} \text{Duke} & \text{Miami} & \text{UNC} & \text{UVA} & \text{VT} \end{matrix} \\ \begin{matrix} \text{Duke} \\ \text{Miami} \\ \text{UNC} \\ \text{UVA} \\ \text{VT} \end{matrix} & \begin{pmatrix} 0 & 0.2617 & 0.2414 & 0.2788 & 0.2182 \\ 0.2094 & 0 & 0.3215 & 0.3055 & 0.1636 \\ 0.2439 & 0.3299 & 0 & 0.1578 & 0.2684 \\ 0.1637 & 0.2054 & 0.1750 & 0 & 0.4559 \\ 0.1005 & 0.4653 & 0.1863 & 0.2479 & 0 \end{pmatrix} \end{matrix} \quad \text{and} \quad \mathbf{r} = \begin{pmatrix} .15 \\ .24 \\ .19 \\ .20 \\ .21 \end{pmatrix}$$

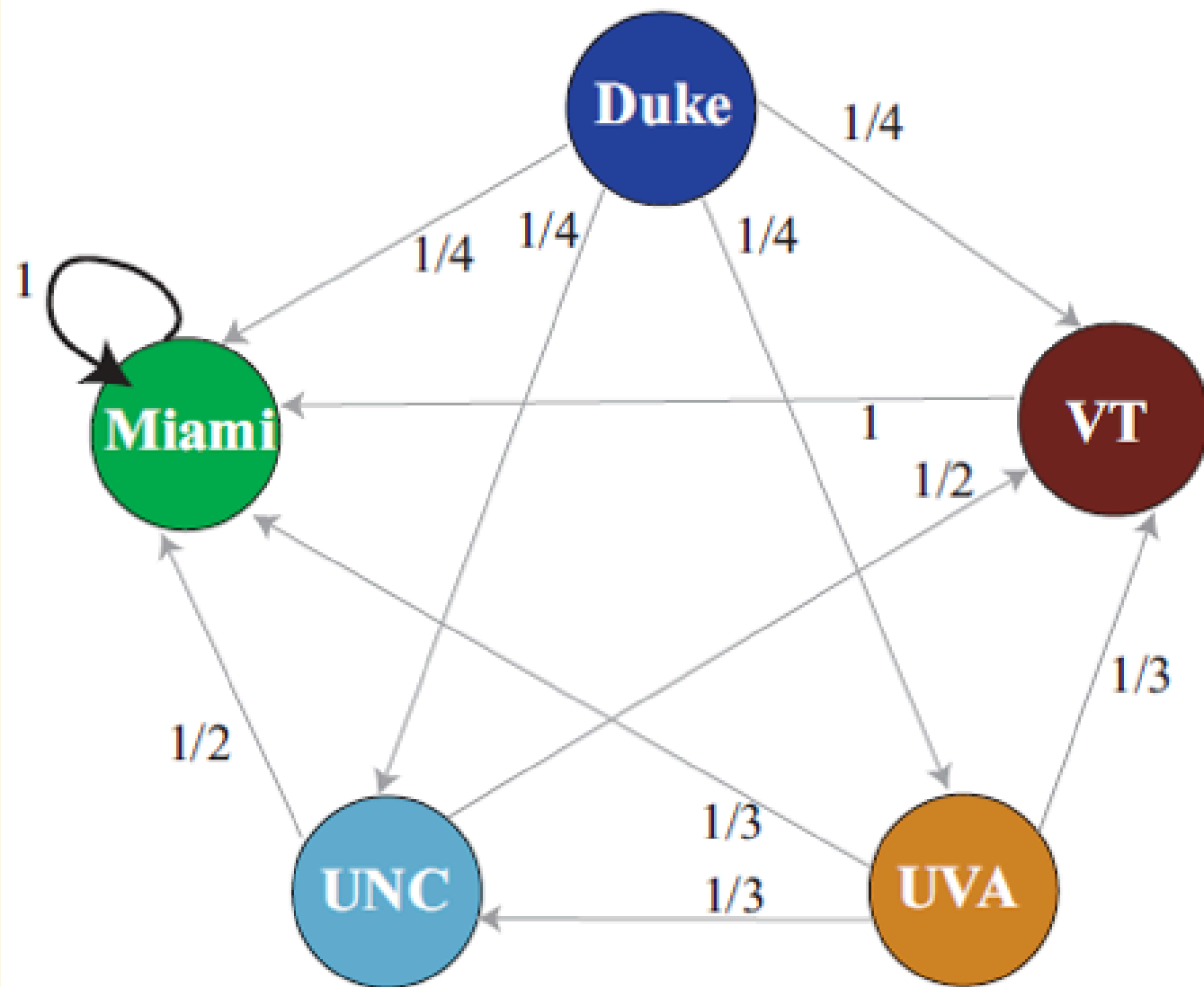
(rating compression starts manifesting)

ISSUES AT THE EXTREMES

HANDLING UNDEFEATED TEAMS



undefeated teams vote
equally for all teams



undefeated teams vote
only for themselves

A random walker soon get stuck with Miami!

Assign a probability to escape:

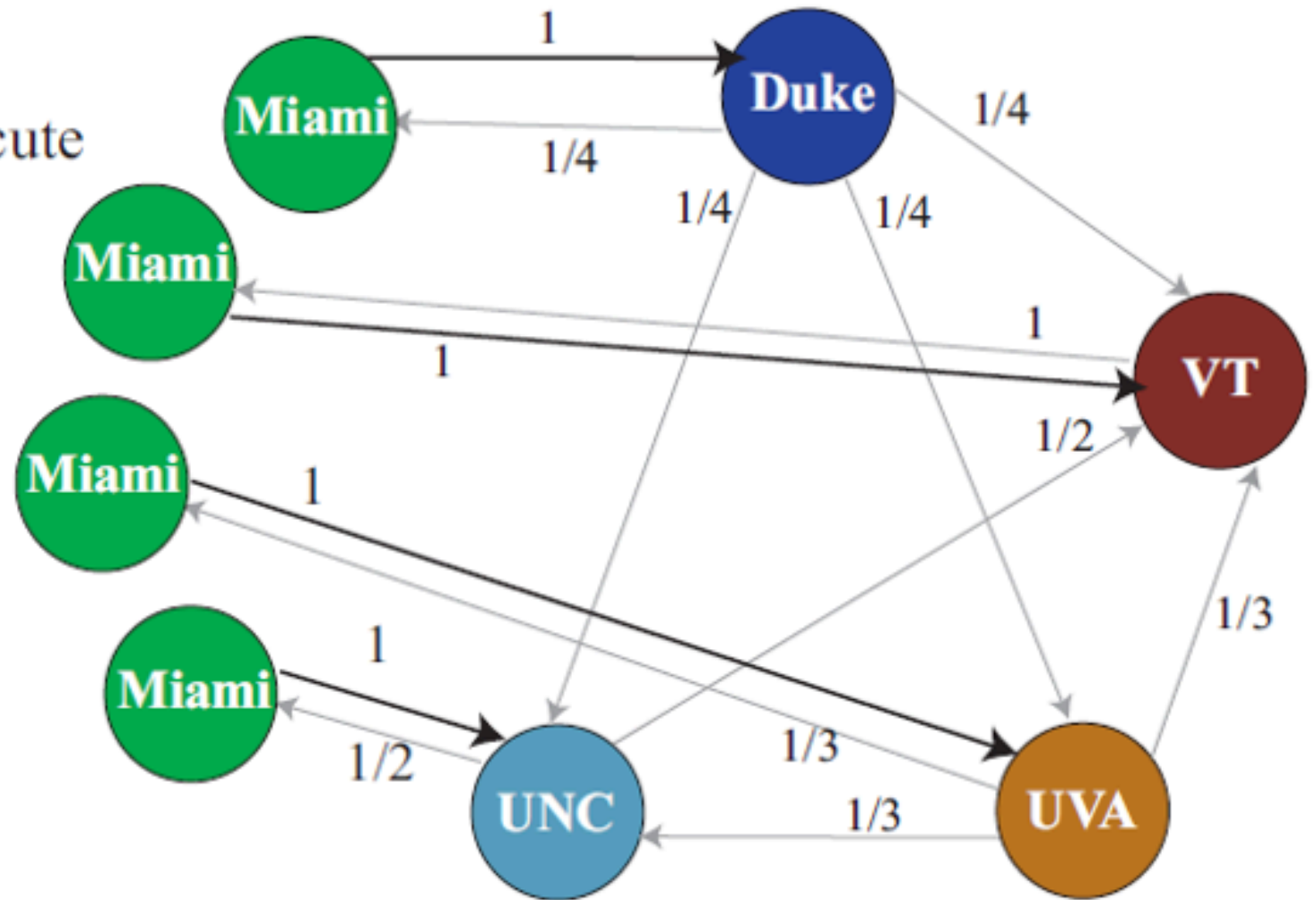
$$\bar{\mathbf{S}} = \beta \mathbf{S} + (1 - \beta) \left[\frac{1}{n} \mathbf{1}^{\mathbf{T}} \mathbf{1} \right] \text{ (a matrix with } \frac{1}{n} \text{ everywhere)}$$

PageRank: $\beta = 0.85$

NFL: $\beta = 0.6$

NCAA: $\beta = 0.5$

undefeated teams execute
a bounceback vote



A better example: modeling the 'Back' button of the browser when we visit a dead-end page.

SUMMARY OF THE METHOD

THE ALGORITHM

MARKOV METHOD FOR RATING TEAMS

1. Form $\mathbf{S}$ using voting matrices for the k game statistics of interest.

$$\mathbf{S} = \alpha_1 \mathbf{S}_{stat1} + \alpha_2 \mathbf{S}_{stat2} + \dots + \alpha_k \mathbf{S}_{statk},$$

where $\alpha_i \geq 0$ and $\sum_{i=1}^k \alpha_i = 1$.

2. Compute $\mathbf{r}$, the stationary vector or dominant eigenvector of $\mathbf{S}$. (If $\mathbf{S}$ is reducible, use the irreducible $\bar{\mathbf{S}} = \beta \mathbf{S} + (1 - \beta)/n \mathbf{E}$ instead, where $0 < \beta < 1$.)

COMPARISON WITH MASSEY'S

The point-differential M. chain:

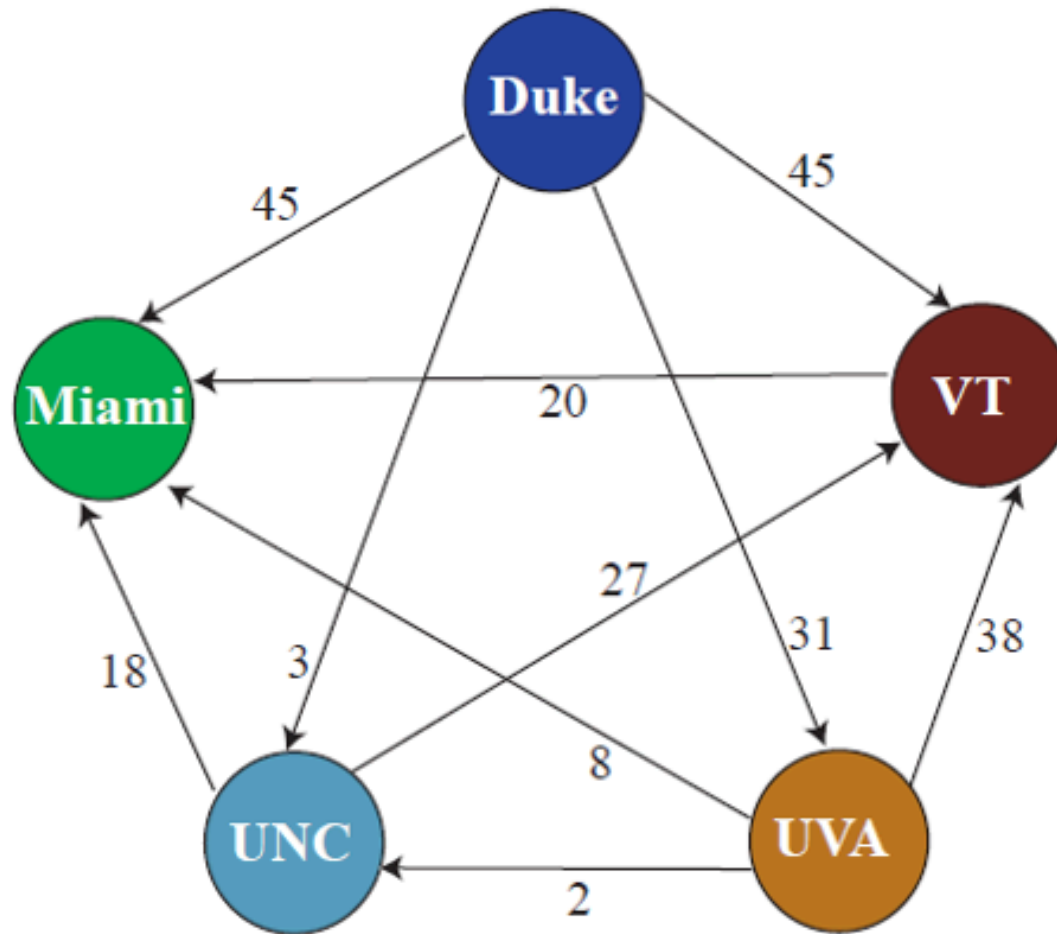


Figure 6.2 Markov graph of point differential voting matrix

Massey graph for the same season

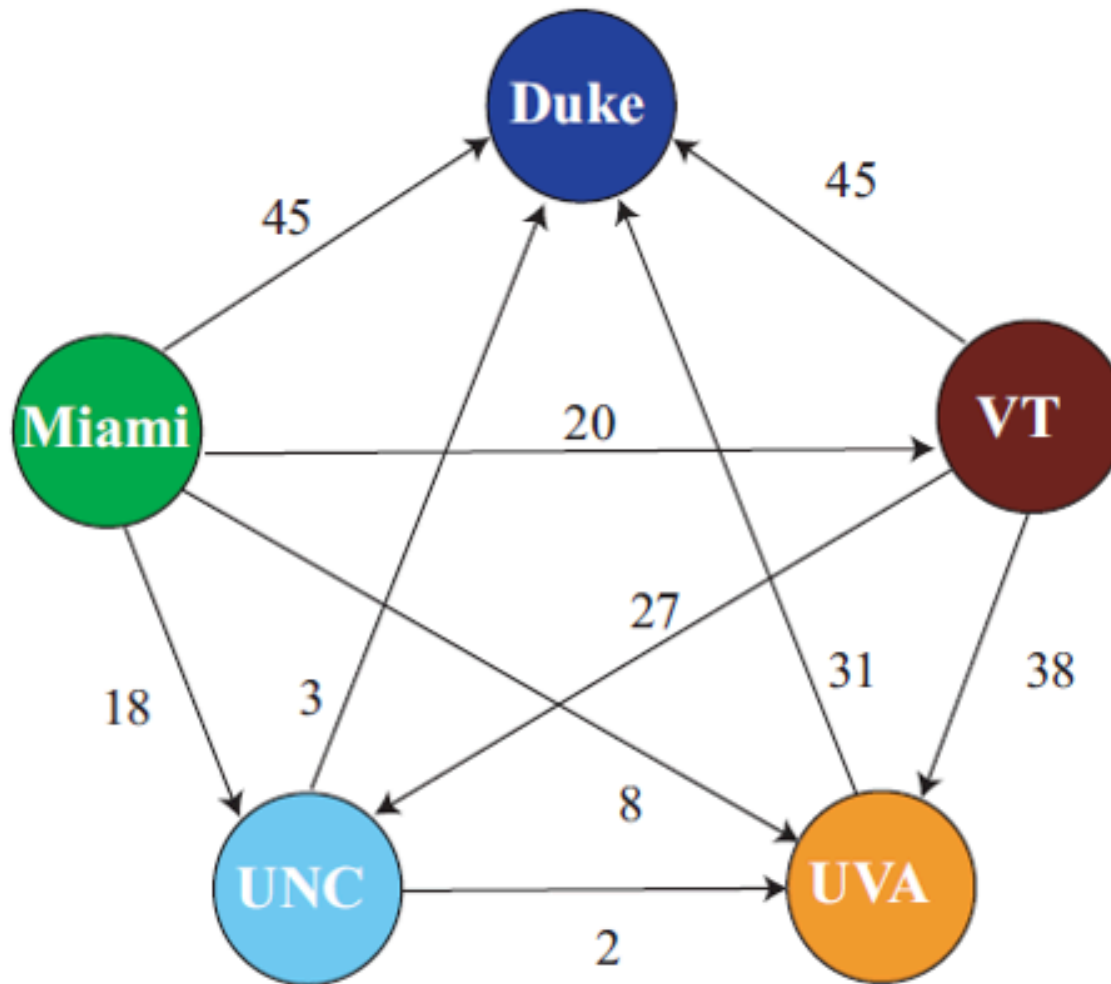


Figure 6.3 Massey graph for the same five team example as Figure 6.2

FURTHER APPLICATIONS

Let's hire fairweather fans to do random walks:

by accumulation and stabilisation of the frequencies we will find out the dominant e-vector of $\mathbf{S}$ *without engaging in matrix operations*.

CODA: RANDOM WALKS FOR MACHINE VISION

IMAGE SEGMENTATION

(a)



Find objects inside a picture

Could random walker discover the perimeter of objects by walking *around* them?

THE DATA

A photos (bitmap) can be seen as

- a $m \times n$ matrix, each value, the pixel being an RGB encoding over $[0..255]$
- a $m \times n \times 3$ tensor where each layer, sometimes called *channel* contain $[0..255]$ intensities of the respective color

a network of pixel nodes joint in a mesh: each node is connected rectilinearly with 2 (corner), 3 (border) or 4 (inner) neighbor pixels.

MAPPING

RGB values can be normalised to $[0..1]$ by mapping the three values into *intensities*, i.e, the length of the vector over N^3

- total black: $[0][0][0] \rightarrow 0$
- total white: $[255][255][255] \rightarrow 1$
- total red: $[255][0][0] \rightarrow ?$

The normalised norm:

$$|p_{ij}|_3 = \frac{1}{\sqrt{3}} \sqrt{\frac{p_{ij}^{red} + p_{ij}^{green} + p_{ij}^{blue}}{255}}$$

THE NORM IN ACTION

So, for a total-red pixel:

$$|p_{ij}|_3 = \frac{1}{\sqrt{3}} \sqrt{\frac{255+0+0}{255}} = \frac{1}{\sqrt{3}} \approx \frac{1}{1.732} \approx 0.57735.$$

For a total-brown pixel:

$$|p_{ij}|_3 = \frac{1}{\sqrt{3}} \sqrt{\frac{255+0+255}{255}} = \frac{1}{\sqrt{3}} \sqrt{2} \approx \frac{1.4142}{1.732} \approx 0.8165.$$

THE RANDOM WALK MODEL

Let random walkers to prefer to remain on the same likely surface/object, i.e., not cross-through density *slopes*

make the prob. to move to a neighbor pixel inverse-proportional to the difference in intensity between the origin and destination pixels.

